ZHEN XU

Ph.D. Candidate, Columbia University

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EDUCATION

Columbia University *2027 (Expected)*
Ph.D. in Cognitive Science

Columbia University *2024*
M.S. in Learning Analytics

Beijing Normal University *2022*
M.Ed. in STEM Curriculum and Teaching Pedagogy

Beijing University of Chemical Technology *2019*
B.E. in Mechanical Engineering

RESEARCH INTERN & APPLIED EXPERIENCE

ETS *2026 Summer*
Research Intern

MatrAIx *2026 Summer*
Research Collaborator

PUBLICATIONS

† equal contribution

Too smart for its own good? Competence Leakage in LLM Simulations of Young Learners
Zhen Xu, Jamie N. Mikeska, Beata Beigman Klebanov. [🔗][Under submission]

AgentPersonaBench: Benchmarking Persona-Driven User Simulation
Jintao Huang, Yifan Wang, Hongyu Shen, . . . , Zhen Xu, et al.
arXiv preprint arXiv:2610.04379 [🔗][Under submission]

Asynchronous Discussion Forums Show More Analytical but Less Diverse Student Engagement in the Age of AI
Zhen Xu, Y. Dai, C. Shi, S. Li, C. Yao, R. Yu.
Proceedings of the Thirteenth ACM Conference on Learning@Scale (2026), 563–566. [🔗] [L@S'26]

Enhancing LLM-Based Data Annotation with Error Decomposition.
Zhen Xu, Vedant Khatri, Yijun Dai, Xiner Liu, Siyan Li, Xuanming Zhang, Renzhe Yu.
The 16th International Learning Analytics & Knowledge Conference (2026) [🔗] [LAK'26]

- **Best Paper Nomination**

From Leaderboards to Decisions: A Psychometric Perspective on Criterion-Referenced Evaluation of Large Language Models
Chengyuan Yao, Zhen Xu,Jiayu Zheng, Renzhe Yu.
LAK26' Workshop: LLM Psychometrics (2026) [🔗][LAK'26 Workshop]

Collaborative Multi-Agent Test-Time Reinforcement Learning for Reasoning.
Zhiyuan Hu, Yunhai Hu, Juncheng Liu, Shuyue Stella Li, Yucheng Wang, Zhen Xu et al.
arXiv preprint arXiv:2601.09667 (2026) [🔗][Under submission]

From Diagnosis to Redesign: Using Quantitative Ethnography to Improve Multi-Agent LLM Reasoning
Vedant Khatri, Anthony Cusimano, Zachari Swiecki, Zhen Xu, Xiner Liu, Renzhe Yu
Advances in Quantitative Ethnography. ICQE 2026. [🔗] [ICQE'26]

How Does Social Annotation Develop Over Time? Examining Temporal Changes in Discourse Moves During Online Reading Activities
Xiner Liu, Yuwei Liang, Zhanlan Wei1, Zhen Xu, Bodong Chen, Ryan S. Baker, Yunseo Lee, Amanda Barany
Advances in Quantitative Ethnography. ICQE 2026. [🔗] [ICQE'26]

Calibrating Social Bias in LLM and Human Responses in Postsecondary Online Discussion Forums
Daniel March, Zhen Xu,Chengyuan Yao, Renzhe Yu.
The thirteenth ACM Conference on Learning @ Scale (2026) [🔗] [L@S'26]

Asynchronous Discussion Forums Show More Analytical but Less Diverse Student Engagement in the Age of AI
Zhen Xu, Yijun Dai, Chenxi Shi, Siyan Li, Chengyuan Yao, Renzhe Yu.
The thirteenth ACM Conference on Learning @ Scale (2026) [🔗] [L@S'26]

Evaluating 21st-Century Competencies in Higher Education Curricula with Large Language Models: Benchmarking and Targeted Prompting Strategies.
Zhen Xu, Xin Guan, Chenxi Shi, Qinhao Chen, Renzhe Yu.
Journal of Learning Analytics, 1–24 (2026) [🔗] [Journal]

Bringing Pedagogy into Focus: Evaluating Virtual Teaching Assistants in Asynchronous Learning Environments.
†Siyan Li, †Zhen Xu, Vethavikashini Chithrra Raghuram, Xuanming Zhang, Renzhe Yu, Zhou Yu.
Findings of the Association for Computational Linguistics: EMNLP (2025) [🔗] [EMNLP'25]

From Course to Skill: Evaluating LLM Performance in Curricular Analytics.
Zhen Xu, Xinjin Li, Yingqi Huan, Veronica Minaya, Renzhe Yu.
The 26th International Conference on Artificial Intelligence in Education (2025) [🔗] [AIED'25]

Whose ChatGPT? Unveiling real-world educational inequalities introduced by LLMs.
Renzhe Yu, Zhen Xu, Sky CH-Wang, Richard Arum.
URL: arXiv preprint arXiv:2410.22282. (2024) [🔗] [Preprint]

Contexts Matter but How? Course-Level Correlates of Performance and Fairness Shift in Predictive Model Transfer.
†Zhen Xu, †Joseph Olson, Nicole Pochinki, Jason Zheng, Renzhe Yu.
The 14th Learning Analytics and Knowledge Conference, 713–724 (2024) [🔗] [LAK'24]

Theory, Research, and Practice of Peer Tutoring in Higher Institutions Worldwide.
Zhen Xu, Ping Zhang, Lin Ding, Wenkai Zhang.
Physics and Engineering, 9–12 (2023) [In Chinese] [🔗] [Journal]

Analysis of Funding for Physics Education Research by the National Science Foundation of the United States Between 2011 and 2020.
Ping Zhang, Ding Lin, Zhen Xu.
Science Foundation of China, 516–522 (2022) [In Chinese] [🔗] [Journal]

Example of Research-Based Physics Curriculum Development with Funding Support.
Zhen Xu, Ping Zhang, Lin Ding.
Physics and Engineering, 72–75 (2021) [In Chinese] [🔗] [Journal]

Experimental/Quasi-Experimental Research in Physics Education Research.
Lin Ding, Ping Zhang, Zhen Xu.
Physics and Engineering, 3–7 (2021) [In Chinese] [🔗] [Journal]

SELECTED POSTERS, PRESENTATIONS, AND WORKSHOPS

How Have Instructors Adapted to Generative AI? Evidence from 90,000 Assignments in Postsecondary Courses.
†Jinwon Kim, †Zhen Xu, Renzhe Yu.
Stanford Education Data Science Conference 2026 (2026) [EDS'26]

- **Best Full Paper Award**

Intersectional inequalities in asynchronous academic communication: Before and after the introduction of GenAI.
†Zhen Xu, †Sky CH-Wang, Zhou Yu, Renzhe Yu.
The 54th Annual Meeting of the Society for Computation in Psychology (2024) [SCiP'24]

- **Best Student Paper Award**

Applying Visual Data Pattern Analytics to Public NYC District Equity Indicators Data.
Alex J. Bowers, Zhen Xu, Feiyi Sun.
The Annual Meeting of the University Council for Educational Administration (2024) [UCEA'24]

Intersectional Inequalities in Online Communication Among Post-secondary Learners.
Renzhe Yu, Sky CH-Wang, Zhen Xu, Zhuojian Wei.
The 9th International Conference on Computational Social Science (2023) [IC2S2'23]

INVITED TALKS AND PANELS

Guest Speaker and Panelist: Educational Technology Sub-Forum of the 16th China Education Symposium at Harvard University 2025

Guest Speaker: AI Application in the Educational Sub-Forum of the Columbia University Teachers College China-U.S. Education Forum 2025

TEACHING EXPERIENCE

Teachers College, Columbia University Spring 2024
· HUDK 4051: Learning Analytics: Process and Theory (TA)
· HUDK 4051: Learning Analytics: Process and Theory: Coding Lab Sessions for LLM (Lecturer)

Beijing No. 4 High School Fall 2021
· Science and Physics (Lecturer)

Beijing Normal University Spring 2021
· Electromagnetism (TA)
· Nature of Science (TA)

COMMUNITY SERVICE

Organizer

- **Workshop Co-Organizer:** Curriculum Analytics for 21st-Century Competencies in Postsecondary Education. The 16th International Learning Analytics & Knowledge Conference, Bergen, Norway

Conference and Professional Service

- **PC member:** The 27th International Conference on Artificial Intelligence in Education (AIED 2026); 2nd AI Literacy for All (AIED 2026); The 17th Learning Analytics and Knowledge Conference (LAK 2027)
- **Co-Chair:** Doctoral Student Forum, The Global Chinese Conference on Computers in Education (GCCCE 2026)
- **E-Board member:** Data Science and Education Association, Teachers College, Columbia University

Reviewer

- The First Workshop on Interpreting Agent Behavior(NeurIPS 2026)
- The 27th International Conference on Artificial Intelligence in Education (AIED 2026)
- The Internet and Higher Education (Journal)
- ACM Conference on Learning at Scale (L@S 2026, L@S 2025)
- The Seventh AAAI/ACM Conference on AI, Ethics, and Society (AIES 2024)
- The 14th Learning Analytics and Knowledge Conference (LAK 2024)

HONORS AND SCHOLARSHIPS

Ben Wood Fellowship at Teachers College, Columbia University	2024
First-class scholarship for graduate study at Beijing Normal University	2021
Outstanding Graduates of Beijing (Top 5 across the major cohort)	2019
National Scholarship (Top 1 in the major cohort)	2017